

AI, Human Cognition and Knowledge Collapse

Acemoglu, Kong & Ozdaglar (2026)

William Gradost

Artificial Intelligence and Economic Modeling · Repository 4

<https://github.com/wbgradost/ai-04-acemoglu>

1 · The paper and the agent's problem

Question. Can accurate agentic AI improve today's decisions while eroding the human learning that sustains tomorrow's collective knowledge?

A short-lived, atomistic agent chooses $e \geq 0$, taking public precision X_t and agentic precision τ_A as given. Effort produces both context-specific and “thin” public information, but the latter is an externality.

$$Y = \sigma^{-2} + \lambda_I e + \tau_A \quad (4)$$

Under Assumption 1 ($\Delta_I = 0$, $\Delta_X > 0$),

$$\max_{e \geq 0} f(0, 0) + G(X_t)\Delta_G + G(X_t)G(Y)\Delta_X - \frac{\varepsilon}{\varepsilon + 1} e^{(\varepsilon+1)/\varepsilon}. \quad (6)$$

$G(\tau) = 2\Phi(\sqrt{\tau}) - 1$: probability that a Gaussian prediction error with precision τ lies within the success band.

Observation 1 (pp. 14–15). For positive precisions, $\lambda_I > 0$, and Assumption 1,

$$U_e = \Delta_X G(X_t) \lambda_I g(Y) - e^{1/\varepsilon},$$

$$U_{eX} = \Delta_X \lambda_I g(X_t) g(Y) > 0, \quad U_{e\tau_A} = \Delta_X G(X_t) \lambda_I g'(Y) < 0,$$

since $g = G' > 0$ and $g'(\tau) = -\frac{1}{2}(1 + \tau^{-1})g(\tau) < 0$.

Mechanism. General knowledge makes context-specific learning useful; AI supplies context-specific precision directly, where returns are diminishing.

Dynamic conditions. If $\varepsilon < 4$, every $X_1 > 0$ converges to the high state (Proposition 3). If $\varepsilon > 4$ and $\tau_A < \tau_A^c$, collapse and the high state coexist and X_1 selects the basin; if $\tau_A > \tau_A^c$, collapse is unique and globally stable (Proposition 5). The boundary $\varepsilon = 4$ is not characterized.

3 · What I did

- ▶ Reconstructed equations (3)–(6) and the effort FOC from the May 5, 2026 MIT copy (69 pages), keeping equation and page references.
- ▶ Derived both cross-partial from $g(\tau) = (2\pi)^{-1/2}e^{-\tau/2}\tau^{-1/2}$, including the corner $X_t = 0 \Rightarrow e = 0$.
- ▶ Traced the feedback loop:

$$\tau_A \uparrow \Rightarrow e_t \downarrow \Rightarrow X_{t+1} \downarrow \Rightarrow e_{t+1} \downarrow .$$

- ▶ Audited welfare: under Assumption 2, $\sigma^{-2} \geq \sqrt{2} - 1$, Propositions 10–11 give a finite threshold $\tau_A^* \geq 0$: welfare rises before it (if that interval is nonempty) and falls after it.
- ▶ Checked Section 5 and the appendix before proposing an extension: aggregation, synthetic data, and effort separability are already covered; the production restriction $\Delta_I = 0$ is not relaxed.

4 · Where I did not believe the AI

AI claim audited: “Public precision complements effort, while agentic-AI precision substitutes for effort.”

Check by hand. Starting from equations (4) and (6), I derived the effort FOC and then differentiated the marginal payoff:

$$U_{eX} = \Delta_X \lambda_I g'(X_t) g(Y) > 0,$$

$$U_{e\tau_A} = \Delta_X G(X_t) \lambda_I g'(Y) < 0,$$

because $g'(z) = -(z+1)\phi(\sqrt{z})/(2z^{3/2}) < 0$.

Verdict: confirmed, with a boundary. The strict signs hold for positive precisions under Assumption 1; at $X_t = 0$, optimal effort is the corner $e = 0$ (Observation 1, pp. 14–15).

Observation 1

$$Y_{it} = \phi^{-2} + \lambda_I e_{it} + \tau_A$$

- $\frac{\partial Y}{\partial e} = \lambda_I$
- $\frac{\partial Y}{\partial \tau_A} = 1$

Ut relevante: $U_{it} = f(0,0) + G(X_t) \Delta_0 + G(X_t) G(Y_{it}) \Delta_X - \left(\frac{\epsilon}{\epsilon+1}\right) e_{it}^{\frac{\epsilon}{\epsilon+1}}$

$$\Delta_X = 0, \Delta_X > 0, \lambda_I > 0, g(z) = G'(z) > 0$$

$$\frac{\partial U_{it}}{\partial e_{it}} = G(X_t) \Delta_X G'(Y_{it}) \frac{\partial Y_{it}}{\partial e_{it}} - \left(\frac{\epsilon}{\epsilon+1}\right) \left(\frac{\epsilon+1}{\epsilon}\right) e_{it}^{\frac{\epsilon}{\epsilon+1}}$$

$$\frac{\partial U_{it}}{\partial e_{it}} = \Delta_X G(X_t) \lambda_I g(Y_{it}) - e_{it}^{\frac{\epsilon}{\epsilon+1}}$$

$$\frac{\partial^2 U_{it}}{\partial e_{it} \partial X_t} = \Delta_X \lambda_I G(X_t) g(Y_{it}) = \Delta_X \lambda_I g(X_t) g(Y_{it}) > 0$$

→ X_t y e_{it} son complementarios

$$\frac{\partial^2 U_{it}}{\partial e_{it} \partial \tau_A} = \Delta_X G(X_t) \lambda_I g'(Y_{it}) \frac{\partial Y_{it}}{\partial \tau_A} = \Delta_X G(X_t) \lambda_I g'(Y_{it}) < 0$$

→ τ_A y e_{it} son sustitutos para $X_t > 0$

...

$$G(z) = 2\Phi(\sqrt{z}) - 1$$

$$g(z) = \frac{\phi(\sqrt{z})}{\sqrt{z}}$$

$$g'(z) = \frac{d}{dz} \left[\phi(\sqrt{z}) z^{-1/2} \right] = -\frac{\phi(\sqrt{z})}{2\sqrt{z}} - \frac{\phi(\sqrt{z})}{2z^{3/2}} = -\frac{(z+1)\phi(\sqrt{z})}{2z^{3/2}} < 0, z > 0$$